ENGINEERING PORTFOLIO

# Rio Ben-Yehoshua

Mechanical engineering student at The Cooper Union in New York City, graduating May 2029. **Looking for a Summer 2027 internship in mechanical design, controls and mechatronics.**

GPA 3.77 | Dean's List | Half-Tuition Scholarship


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## ABOUT

I'm a sophomore studying mechanical engineering. I like designing and building systems that actually work: build it, test it, see where it fails, and fix it. I'm most interested in controls and mechatronics, where mechanical design meets sensors, motors and code.

English and Hebrew (fluent), Chinese (conversational).

## SKILLS

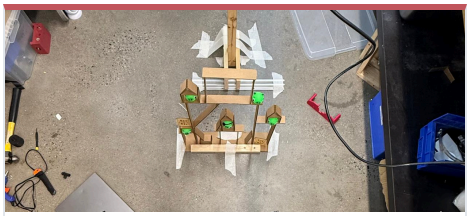
**PROGRAMMING**
C, Python, Arduino

**CAD & TOOLS**
SolidWorks, Fusion, Onshape, LaTeX, Excel

**ELECTRONICS**
L298N motor drivers, MOSFET switching, load cells + HX711, I2C displays, Arduino, ESP32

**FABRICATION**
3D printing, laser cutting, soldering, band saw, table saw; machine shop certified

## PROJECTS IN THIS PORTFOLIO



110 PASSES IN 60 S · 1ST IN CLASS

### Ping Pong Recirculation System

Autonomous launch-and-recover machine. Page 2.



0.84% FULL-SCALE ERROR

### Digital Load Cell Scale

Four load cells, HX711, Arduino. Page 3.



72 H SELF-WATERING · 0 PUMPS

### PlantPlex

Self-watering indoor farming grid. Page 4.

● IN PROGRESS · SINCE OCT 2026

### Crazyflie drone swarm

With a professor: programming Crazyflie nano-drones in Python to fly coordinated formations, with physics-model motions (orbits, mass-spring) a possible next step. Calibrated the Lighthouse positioning.

● IN PROGRESS · SINCE SPRING 2026

### Cooper Union Motorsports (FSAE)

Body Cockpit Safety team. Working on the pedal box: updating my SolidWorks model to match the real assembly, building parts, and now fitting and mounting the master cylinder.

## WORKSHOP BUILDS



### Custom cardboard chair for a preschooler

For a child with cerebral palsy who often fell out of his seat. Triwall cardboard with laser-cut sides, arm supports, a sliding foot stool and a storage shelf.



### Sand casting a dog-bone specimen

Sand cast 6061 aluminum, tensile tested to failure on an Instron. Peak force 8.90 kN; fractured at about 1.9% strain.

EID-103 PRINCIPLES OF DESIGN · SPRING 2026

# Ping Pong Recirculation System

An autonomous launch-and-recover machine: fire six balls through a 6″ × 6″ hoop, catch them, reload them, and repeat for 60 seconds with no human touch after one button press.

**110**
PASSES IN 60 S

**1st**
PLACE IN CLASS

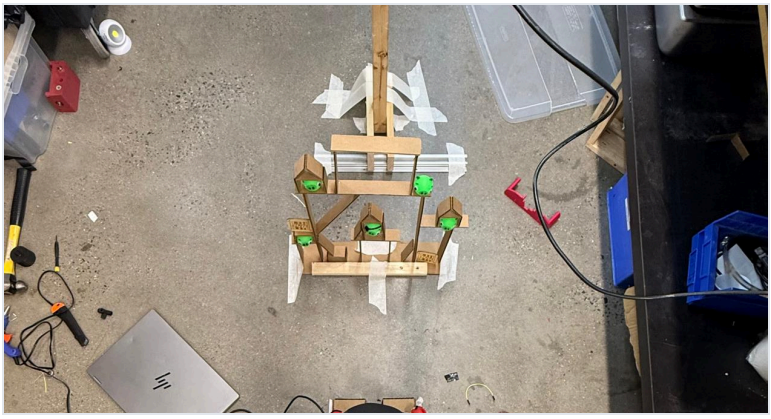
**±5%**
IR HOOP SPEED ACCURACY

MY ROLE
Launcher, return loop, launcher control code

TEAM
4 students

TIMELINE
Mar to May 2026

TOOLS
Onshape, 3D printing, laser cutting, Arduino



The full system, top-down: launcher, hoop, backboard and catch net.

## WHAT I DID

- Designed and 3D-printed the launcher mounts and funnel through many revisions
- Iterated the catch net and its reinforcement through repeated testing
- Wrote the launcher control code: two flywheel motors on an L298N driver, a MOSFET-switched funnel motor, one-button start
- Built an 8-sensor IR hoop that measured ball speed in m/s to within ±5%, checked against balls dropped from rest 1 ft above it (free fall: about 2.4 m/s)

## DESIGN DECISIONS

### Open return, not a tube

Other teams used enclosed tubes. Ours rebounds off a curved backboard into a net and a funnel feeds the wheels. Riskier, but it removed the slowest part of recirculation.

### Two-wheel flywheel

Two printed wheels with a grip surface pinch each ball from both sides. Speed approximately 5 m/s, measured by our camera system. The funnel starts 2.5 s after the wheels so they are at speed first.

### Speed from two planes

The IR hoop timed each ball between two planes of four beams, using hardware interrupts for microsecond timestamps and a 200 ms lockout so no ball counted twice.

## WHAT BROKE, AND WHAT FIXED IT

**Motors shook the mounts apart.** Added washers for smoother spinning and kept iterating the mount until it survived full runs.

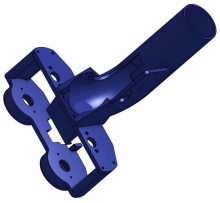
**Wheel grip kept failing.** Rubber bands snapped with repeated use; a superglued high-grip material held.

**Balls escaped the net.** Switched to a net with slack to absorb energy, and reinforced the spots where testing showed escapes.

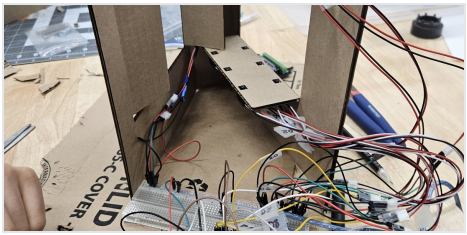
**Jams in the funnel.** A small feed motor helped; once launches were fast enough, balls never queued up.



The final launcher.



Launcher mount and funnel (CAD).



The 8-sensor IR hoop prototype.

ME 104 · SPRING 2026

# Digital Load Cell Scale

A digital scale built from four load cells, an HX711 amplifier and an Arduino, in a layered 3D-printed enclosure with an LCD, a tare button and gram/ounce switching.

**0.84%** FULL-SCALE ERROR

**$R^2$ = 0.9999** SLOPE 0.997

**±0.6 g** STD. DEV., 10 READINGS

MY ROLE
Enclosure CAD and the weighing code

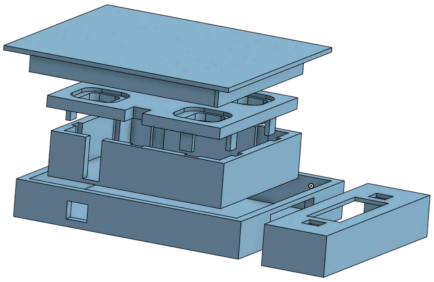
TEAM
3 students

TIMELINE
Apr to May 2026

TOOLS
CAD, 3D printing, Arduino, HX711, I2C LCD



The finished scale.



Exploded view: platform, load cell frame, base and front box.

### Four load cells, not one

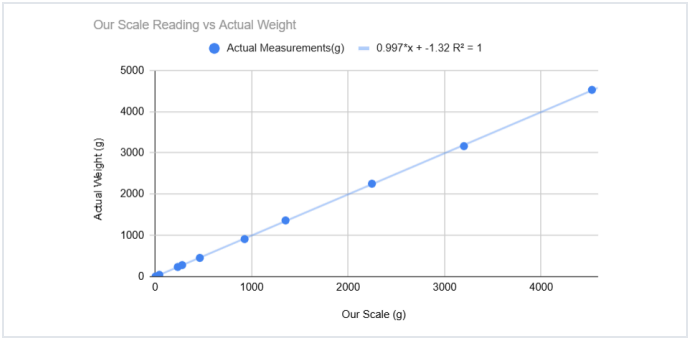
Like a bathroom scale: a cell near each corner keeps a larger platform level and reduces error from off-center loads. Wired together, they form a full Wheatstone bridge.

### Layers with one job each

Platform, load cell frame and electronics bay are separate layers, so the electronics are protected from load and movement and the cells stay aligned.

### Averaging, and a trade-off

Each reading shown is a fresh average of the latest batch, about once a second. A running average would look smoother but blends in old values; a fresh batch responds as soon as the load changes.




Scale reading vs. actual mass: y = 0.997x − 1.32.

TEST RESULTS (TEN KNOWN MASSES)

| Actual (g) | Scale (g) | Error |
|---|---|---|
| 4 | 3.9 | 2.50% |
| 44 | 43.2 | 1.82% |
| 234 | 234.1 | 0.04% |
| 280 | 279.8 | 0.07% |
| 454 | 463.1 | 2.00% |
| 910 | 926.9 | 1.86% |
| 1362 | 1351.2 | 0.79% |
| 2252 | 2246.7 | 0.24% |
| 3162 | 3200 | 1.20% |
| 4526 | 4526.7 | 0.02% |

Max tested load 4535.9 g (10 lb), capped to protect the printed platform. Percent error is largest at small masses, as expected.

**If I rebuilt it:** replace the blocking delays in the button code with non-blocking timing so the display never pauses, and track down the slow reading drift (likely signal interference or load cell sensitivity).

EID 101-D · FALL 2025

# PlantPlex

A freestanding, double-sided grid of removable planter pods that wick water up to the plants by capillary action, so a small apartment can grow its own produce with no pump and no gardening experience.

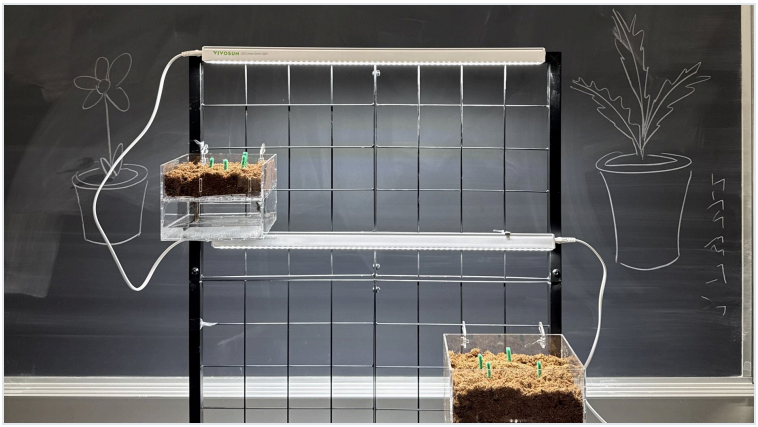
| 72 h | 0 | 24 | 24″ × 24″ |
|---|---|---|---|
| SELF-WATERING TEST | PUMPS | REMOVABLE PODS | FOOTPRINT |

**MY ROLE**
Pod, grid and wicking design; built the prototype

**TEAM**
7 students

**TIMELINE**
Sep to Dec 2025

**TOOLS**
Rapid prototyping, capillary wicking, LED grow lights



The grid prototype with planter pods.

## WHAT I DID

- Designed and prototyped the planter pods, the grid and the wicking system across three iterations
- Built and assembled the working prototype by hand, into a freestanding system that wicks water to every pod
- Co-wrote the final report

## TESTING

- **Watering:** a pod held steady soil moisture for 72 h with no daily watering and no over-saturation
- **Stability:** fully loaded on both sides, the grid showed no significant tilt
- **Lighting:** 60 W LED checked against published light needs for each crop

## THREE VERSIONS, EACH FIXING THE LAST

### 1. Proof of concept

A small self-watering microgreen box. Wall, floor and ceiling mounts were ruled out for a self-supported grid that pods hang from.

### 2. Capillary pods

A wicking column and water reservoir in each pod. Manual refills chosen over a pump: no noise, energy or water cost, and fully modular parts.

### 3. Final design

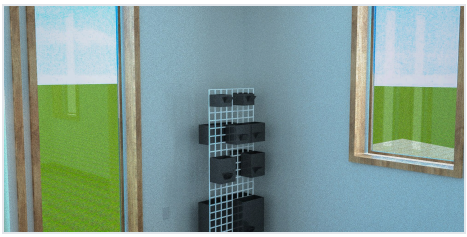
A double-sided grid of 24 modular pods that can be removed, replaced or moved around the grid for customization and easy upkeep, half on each side to balance the weight, with detachable LED lights.

### What we threw out

**One big capillary mat:** heavy, always damp, likely to mold. **Curtain-rod hanging:** blocked daylight, might not hold the weight.



A working pod on the grid.



Rendering in an apartment room.